

DHARAA Smart Irrigation System - Final Design Summary

Objective

DHARAA is a hybrid Physics-Informed and AI-assisted irrigation management system that determines:

1. When to irrigate
 2. How much water to apply
 3. How to safely execute irrigation
 4. How to handle rain uncertainty and emergency crop protection
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Module 1: Soil Moisture Prediction

Purpose

Predict future soil moisture using:

- Current soil moisture
- Temperature
- Humidity
- Crop type
- Soil type
- Season
- Crop age

Output

- Current moisture
- Future moisture prediction
- Date when moisture reaches threshold

This module determines the irrigation trigger date.

Module 2: Irrigation Date and Time Scheduler

Purpose

Select the best irrigation date and time.

Date Selection Window

When threshold moisture is predicted on a specific date:

Example:

Threshold reached on:

- 25 June

Scheduler evaluates:

- 24 June
- 25 June
- 26 June

Three-day scheduling window.

Date Selection Factors

Daily weather forecast:

- Temperature
- Wind speed (highest priority)
- Rain probability
- Weather code

Hard Rejections

Reject date if:

- Rain probability > 80%
- Thunderstorm weather code present

Day Selection Philosophy

Prefer:

- Lower wind
- Moderate temperature
- Later irrigation date when scores are similar

Delay bonus is used to maximize irrigation date.

Time Selection

Hourly weather forecast is used.

Summer

Prefer:

- Evening
- Night

Fallback:

- Early morning

Winter

Prefer:

- Early morning (~5 AM)

Monsoon

Prefer:

- Early morning

Time Selection Factors

- Wind speed (highest priority)
- Temperature
- Rain probability

If no valid time exists:

- Reject that day
- Evaluate next day

Rain Hold Function

Purpose

Delay irrigation when significant rain is expected.

Inputs

- Irrigation date
- Weather forecast CSV
- Irrigation cycle

Logic

Check future rain after selected irrigation date.

If:

Maximum rain probability $\geq$ 80%

Then:

Rain Hold = TRUE

Store:

- Hold date
- Hold reason

Hold Release

On hold date:

If Rain Occurred

- Cancel irrigation
- Recalculate schedule

If Rain Did Not Occur

- Release hold
- Start irrigation

Module 3: Water Requirement Calculator

Purpose

Determine target soil moisture and irrigation requirement.

Inputs

- Current moisture
 - FC
 - Threshold moisture
 - Available water
 - Soil volume
 - Rain forecast
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Initial Target

Target:

FC + Buffer

Buffer:

10-25% of FC

Example:

FC = 30%

Buffer = 5%

Target = 35%

Water Availability Check

Calculate required water.

If:

Available Water < 120% of Required Water

Then:

Remove buffer.

New target:

FC

Rain Adjustment

If future rain probability > 60%:

1. Remove buffer completely
2. Reduce irrigation below FC dynamically

Safety target:

Threshold + 50% of (FC - Threshold)

Example:

FC = 30 Threshold = 20

Safety target = 25

Irrigation never goes below safety target.

Outputs

- Required theta
 - Required water (liter)
 - Water sufficient flag
 - Reason
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Module 4: Emergency Irrigation Monitor

Purpose

Protect crops from severe water stress.

Runs continuously.

Trigger

If:

Current Moisture $\leq$ Threshold - 10%

Example:

Threshold = 20%

Emergency Trigger = 18%

Then:

Emergency Irrigation

Emergency Rules

Ignore:

- Scheduler
- Rain Hold
- Scheduled Date
- Scheduled Time

Only check:

- Device online
- Tank water available
- Current rain sensor
- Current thunderstorm

If safe:

Start irrigation immediately.

Module 5: Irrigation Execution Manager

Purpose

Execute irrigation safely and reliably.

Start Conditions

1. Scheduled time reached
2. Device online
3. Rain hold inactive
4. Tank level above minimum
5. Moisture below target
6. No rain currently

7. No thunderstorm currently

Pulse Irrigation

Pump ON = 4 minutes

Pump OFF = 2 minutes

Purpose:

- Improve infiltration
 - Reduce runoff
 - Stabilize sensors
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Completion Condition

All sensors must reach target.

Sensor1 $\geq$ Required Theta

AND

Sensor2 $\geq$ Required Theta

AND

Sensor3 $\geq$ Required Theta

Then:

Status = COMPLETED

Water Insufficient Handling

If:

water_sufficient = False

Do NOT cancel irrigation.

Send warning.

Continue irrigation until:

- Target reached OR
 - Tank empty
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Stop Conditions

Completed

All 3 sensors reach target.

Partial

- Tank empty
- Rain starts
- Thunderstorm detected

Failed

- Device disconnects
- Safety timeout
- Hardware fault

Cancelled

- Manual stop

Waiting Device

- Device offline before start

Waiting Rain Hold

- Rain hold active

Emergency

- Emergency irrigation executed
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Logging System

For every irrigation event store:

- Scheduled date

- Scheduled time
 - Actual start time
 - Actual end time
 - Moisture before irrigation
 - Moisture after irrigation
 - Final sensor values
 - Required theta
 - Water sufficient status
 - Tank level before
 - Tank level after
 - Rain hold status
 - Emergency trigger status
 - Runtime
 - Status
 - Reason
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Notification System

Notify user when:

- Irrigation started
 - Irrigation completed
 - Irrigation partially completed
 - Tank empty
 - Rain hold activated
 - Rain hold released
 - Emergency irrigation started
 - Device offline
 - Device reconnected
 - Safety timeout
 - Manual stop
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Final DHARAA Architecture

1. Moisture Prediction AI
2. Irrigation Date-Time Scheduler
3. Rain Hold Function
4. Water Requirement Calculator
5. Emergency Irrigation Monitor
6. Irrigation Execution Manager
7. Database Logging
8. User Notification System

The system combines AI prediction, weather-aware scheduling, water-efficient irrigation planning, emergency crop protection, and autonomous execution to maximize water-use efficiency while preventing crop stress and yield loss.